YEAR 9 MATHS · CHAPTER 1 · MIND MAP

Number and financial maths

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1 Integer operations

Multiply or divide: same signs give +, different signs give −.

$4 + (-9) = -5 \quad -6 - (-2) = -4$

$-7 \times (-3) = 21 \quad -24 \div 6 = -4$

Order of operations: $-2 + 3 \times (-4) = -14$

Watch out: $(-3)^2 = 9$ but $-3^2 = -9$

2 Decimal places and significant figures

Look at the next digit. 5 or more rounds up; 4 or less stays.

3.476 to 1 d.p. is **3.5**, to 2 d.p. is **3.48**

0.04082 to 2 s.f. is **0.041** (start at the first non-zero digit)

5683 to 1 s.f. is **6000**

3 Rational numbers

Rational means it can be written as a fraction $\frac{a}{b}$.

$2\frac{2}{5} = \frac{12}{5} = 2.4$ (terminating)

$\frac{2}{3} = 0.666\ldots = 0.\dot{6}$ (recurring)

$0.35 = \frac{35}{100} = \frac{7}{20}$

$\sqrt{2}$ and π are irrational: their decimals never end or repeat.

4 Operations with fractions

Add or subtract: common denominator first. Divide: flip the second fraction and multiply.

$2\frac{1}{3} + 1\frac{3}{4} = \frac{28}{12} + \frac{21}{12} = \frac{49}{12} = 4\frac{1}{12}$

$\frac{3}{5} \div \frac{9}{10} = \frac{3}{5} \times \frac{10}{9} = \frac{2}{3}$

Watch out: $\frac{1}{2} + \frac{1}{3} \neq \frac{2}{5}$. It's $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$

5 Ratios, rates and best buys

Simplify a ratio by dividing both parts by the HCF. A rate compares different units.

$12 : 18 = 2 : 3$

$240 in the ratio 5 : 3: 8 parts, 1 part = $30, so **$150** and **$90**

150 km in 2.5 h = **60 km/h**

Best buy: 500 g for $3.60 is **$0.72** per 100 g; 750 g for $5.10 is **$0.68** per 100 g, so the 750 g pack is better value

6 Percentages and money

Per cent means "out of 100". Divide by 100 to get a decimal.

Fraction to %: $\frac{7}{20} \times 100\% = 35\%$

$12.5\% = 0.125$

Find a percentage: 15% of $80 $= 0.15 \times 80 = \$12$

Find the whole: 30% of an amount is $45, so the amount is $45 \div 0.3 = \$150$

Watch out: check by working backwards: 30% of $150 is $45 ✓

7 Percentage increase and decrease

Use a multiplier: 100% plus the increase, or 100% minus the decrease. Percentage change is always out of the original amount.

Up 12%: $50 \times 1.12 = 56$ Down 15%: $80 \times 0.85 = 68$

% change $= \frac{\text{change}}{\text{original}} \times 100\%$: 40 to 50 is $\frac{10}{40} = 25\%$

8 Profits and discounts

Profit = selling price − cost price. Profit % is out of the cost price.

Cost $40, sold for $52: profit $12, $\frac{12}{40} \times 100\% = 30\%$

Mark up $80 by 25%: $80 \times 1.25 = \$100$

20% off $65: $65 \times 0.8 = \$52$

9 Income

Wage: paid per hour. Salary: a set amount a year. Commission: a % of sales.

$28/h: time and a half **$42/h**, double time **$56/h**

Salary $78 000 a year $\div 52 =$ **$1500** a week

4% commission on $15 000 sales = **$600**

10 The PAYG income tax system

Pay As You Go: your employer takes tax out of each pay and sends it to the ATO.

Taxable income = gross income − deductions

Medicare levy: 2%. On $60 000: $0.02 \times 60\,000 = \$1200$

11 Simple interest

The same interest every year, worked out on the amount you started with.

$I = Prt$: principal, rate as a decimal, time in years

$1500 at 4% p.a. for 3 years: $I = 1500 \times 0.04 \times 3 = \$180$

Total $= \$1500 + \$180 = \$1680$

Watch out: 6 months is $t = 0.5$

12 Compound interest and depreciation

Compound: interest is added each year, then earns interest too. Depreciation: value drops by a % each year.

$A = P(1 + r)^n$ $2000 at 6% p.a. for 2 years: $2000 \times 1.06^2 = \$2247.20$

Depreciation $A = P(1 - r)^n$: a $20 000 car losing 15% a year is worth $20\,000 \times 0.85^2 = \$14\,450$ after 2 years

Interest earned = A − P. Here $247.20, more than simple interest ($240).